

Structure

Duality

Quality

Effects



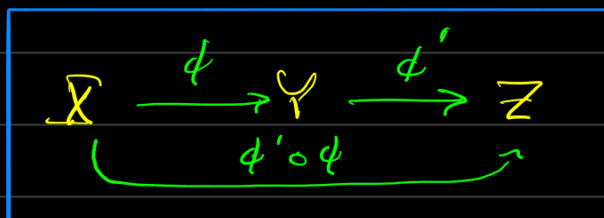
Category

Adjunctions

modalities

monads

→ objective logic



Category: Directed graphs

• Composition is associative with unit 1

Objects: Vertices

Morphism: Edges

* Objects: Think of as generalized spaces such as physical spaces, classifying spaces (moduli spaces ...)

Morphisms: maps between these spaces

or

Objects: data types

morphisms: programs

Groupoid: are sets with gauge symmetry [Generalized symmetries]

ex: Maxwell gauge fields

- object: Gauge configurations
- morph: Gauge trans functions

* Categories are sets with "non-invertible symmetries"

Functor: Maps from one category to another, preserving the (unit &)

composition of morphism

Category

$X \xrightarrow{\phi} Y \xrightarrow{\phi'} Z$

$\searrow \phi' \circ \phi \nearrow$

$\downarrow \quad \downarrow \quad \downarrow$

$\searrow F(\phi' \circ \phi) \nearrow$

functor

$$F(X) \xrightarrow{F(\phi)} F(Y) \xrightarrow{F(\phi')} F(Z)$$



natural
transformations



all squares commute: $G(\phi) \circ \eta(X) = \eta(Y) \circ F(\phi)$

universal
constructors monadic
algebra

adjoints

natural transf

|

functors

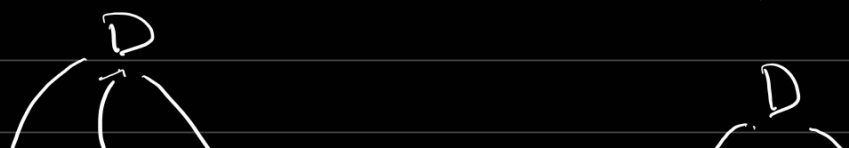
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categories

* Actual "dynamics"/theory comes to be with adjunctions $L \dashv R$

Adjunction: Pair of functors $L: C \rightleftarrows D: R$ and a natural transformation $\eta: id_C \Rightarrow R \circ L$ & $\epsilon: L \circ R \Rightarrow id_D$

satisfying the Yin-Yang identity





ex: $\mathbb{N}$: category

each $i \in \mathbb{N}$: objects

$1 \xrightarrow{\phi} 2$

$\phi \downarrow$
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ϕ : satisfies $a \rightarrow b \Leftrightarrow a < b$

- Partial order on $\mathbb{N}$ regarded as a category
- look at $2\mathbb{N}$ & $(\mathbb{N} - 2\mathbb{N})$:
have functors into $\mathbb{N}$ by embedding

* Initial object: "Empty set" $\emptyset$

terminal object: * "set with 1 element"

- The endofunctors constant on these objects are adjoint $\emptyset \dashv *$

* String Dualities: some are adjoint functors

(i) Double dim reduction

branes in higher dim $\leftrightarrow$ lower dim theory w KK-modes

(ii) T-duality

(iii) $M \leftrightarrow$ Type IIA

Make place for physics: Gros. Toposes:

- Main drive for categories in physics: provide categories of generalized spaces
 $\rightarrow$ field spaces, super-spaces, dg-spaces, moduli spaces, orbifolds, ...
- Done by declaring how these are probed by generalized coord. charts

* These categories are called Topos and is where physics takes place

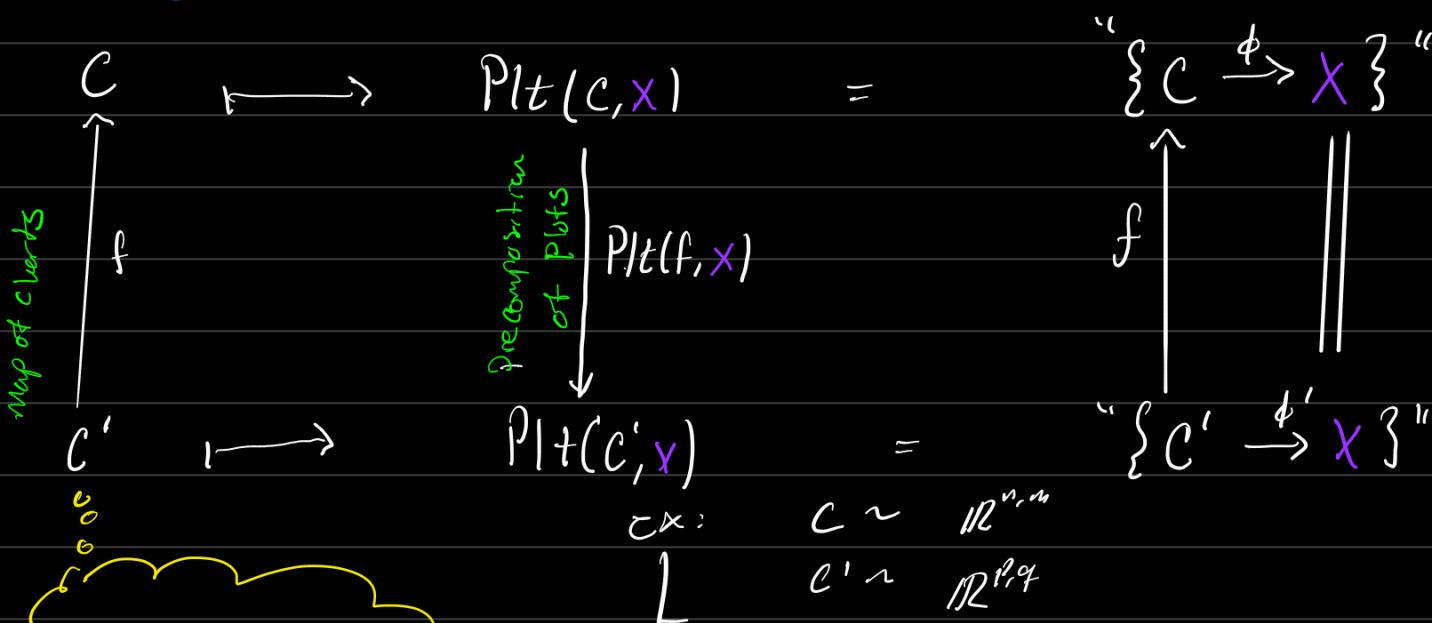
Sheaf: assigns an object (e.g. a set, vector space, or group) to each

U open set of a space and provides a way of "gluing" these

local objects together to form a global object.

- A sheaf $\mathcal{F}$ on a space X assigns to each open set $U \subset X$ an object $\mathcal{F}(U)$ with the restriction map $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$, $V \subseteq U$

generalized charts $\xrightarrow{\text{cover and glue}} \text{plots of } X \text{ to become the maps into } X$



start w/ cat. of generalized charts

Def: A generalized space X is sheaf on generalized charts